

Closing the hook rank formula: the property (Q_m) via strong image and sign induction

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Abstract

The May 9 writeup reduced the hook rank formula $\text{rank } \Pi^{S_n}|_{V_{(n-1,1)}} = \lceil (n-2)/2 \rceil$ to the single property

$$(Q_m) : \quad v_{m-1}^*(K_m) \neq 0 \quad \text{for all even } m \geq 4,$$

where $K_m := \ker \Pi^{S_m}|_{V_{(m-1,1)}}$. Here we prove (Q_m) unconditionally, completing the program.

The proof rests on two pieces. First, a **Strong Image Coefficient Formula**: explicit triangular inversion of the operator $R''_{n-1} = (T_{n-2}+1) \cdots (T_1+1)$ on $V_{(n-1,1)}$ shows that for any w with $R'_n(w) = y \in \text{span}(v_2, \dots, v_{n-2})$,

$$w_{v_l} = \frac{y_{v_{l-1}} - \gamma_{l-2} y_{v_{l-2}}}{(1+q)^{n-2}}, \quad 3 \leq l \leq n-1,$$

where $\gamma_i := \beta_i p_{i+1}$ are explicit Hoefsmit constants. Second, an alternating **sign induction** on the “slopes” μ_l (for l odd) and ν_l (for l even) on the new-generator quotient of K_l . At any positive real q , both slopes are strictly negative, while γ_{m-3} is positive – forcing $\mu_{m-1} - \gamma_{m-3} < 0$, equivalently (Q_m) .

Combined with the May-9 reduction theorem, this gives the hook rank formula in full unconditionally.

1 Setup

We retain all conventions from the May-8 morning, May-8 evening, and May-9 papers (2026-05-08-hook-rank-for-2026-05-08-evening-2-kernel-dichotomy.tex, 2026-05-09-nesting-and-reduction.tex). In particular:

- $V_{(n-1,1)}$ has the seminormal basis $v_2, \dots, v_n$.
- $u_i = v_i + \beta_i v_{i+1}$ is the $+q$ -eigenvector of T_i in our normalization $\alpha_i = 1, q_i = 1/(1+q)$, with

$$\beta_i = \frac{q[i-1]}{[i]}, \quad \beta'_i = -\frac{[i+1]}{[i]}, \quad p_i = \frac{[i+1]}{[2][i]}, \quad q_i = \frac{1}{[2]},$$

where $[k] := (q^k - 1)/(q - 1) = 1 + q + \cdots + q^{k-1}$.

- $K_m := \ker \Pi^{S_m}|_{V_{(m-1,1)}}$, $D_m := R'_m(V_{(m-1,1)}) = \text{span}(v_2, \dots, v_{m-2}, u_{m-1})$.
- (P_m) : $v_m^*(K_m) = 0$ iff m odd.
- (Q_m) : $v_{m-1}^*(K_m) \neq 0$.
- (R_m) : for m even, v_{m-1}^* and v_m^* are proportional on K_m with nonzero ratio.

- Image Lemma (May-8 evening): for $w \in V_{(m-1,1)}$ with $R'_m(w) = y \in \text{span}(v_2, \dots, v_{m-2})$, $w_{v_m} = C_m y_{v_{m-2}}$ with $C_m = \beta_{m-2} \beta'_{m-1} / (1+q)^{m-1}$.
- Nesting (May-9): for all $n \geq 4$, $K_{n-2}^{\text{embed}} \subset K_n$ via the natural embedding.

We define

$$\gamma_i := \beta_i p_{i+1} = \frac{q[i-1][i+2]}{[2][i][i+1]} \quad (i \geq 2),$$

which is positive at every positive real q .

2 The Strong Image Coefficient Formula

Theorem 1 (Strong Image Coefficient Formula). *Let $n \geq 4$ and let $w \in V_{(n-1,1)}$ satisfy $R'_n(w) = y \in \text{span}(v_2, \dots, v_{n-2})$. Modulo $\ker R'_n = \mathbb{Q}(q)v_2$, the coordinate w_{v_l} is uniquely determined for $3 \leq l \leq n$, and*

$$w_{v_l} = \frac{y_{v_{l-1}} - \gamma_{l-2} y_{v_{l-2}}}{(1+q)^{n-2}}, \quad 3 \leq l \leq n-1, \quad (1)$$

with the convention $\gamma_1 = 0, y_{v_1} = 0$. Furthermore, $w_{v_n} = C_n y_{v_{n-2}}$ (the May-8 evening Image Lemma).

Proof. Write $R'_n = (T_{n-1}+1)R''_{n-1}$ with $R''_{n-1} = (T_{n-2}+1) \cdots (T_1+1)$. The image of R''_{n-1} is $I_{n-2} = \text{span}(v_2, \dots, v_{n-3}, u_{n-2}, v_n)$ (Lemma 4 of the May-8 morning paper). The map $R''_{n-1}: V_{(n-1,1)} \rightarrow I_{n-2}$ has 1-dim kernel $\mathbb{Q}(q)v_2$ since $(T_1+1)v_2 = 0$.

Set $z := R''_{n-1}(w) \in I_{n-2}$. We claim that the matrix of R''_{n-1} from the source basis $(v_3, v_4, \dots, v_n)$ (in the quotient $V_{(n-1,1)}/\mathbb{Q}(q)v_2$) to the image basis $(v_2, v_3, \dots, v_{n-3}, u_{n-2}, v_n)$ is lower-triangular with constant nonzero diagonal $(1+q)^{n-3}$.

Computation of the matrix. Let $L_j := (T_j+1)(T_{j-1}+1) \cdots (T_1+1)$. We track $L_j v_l$ for $l \geq 3$ and $j = 1, 2, \dots, n-2$. By induction on j ,

$$L_j v_l = \begin{cases} (1+q)^j q_{l-1} (v_{l-1} + B_{l,l} v_l + B_{l,l+1} v_{l+1} + \cdots + B_{l,j-1} v_{j-1} + B_{l,j} u_j) & \text{if } 3 \leq l \leq j+1, \\ (1+q)^j v_l & \text{if } l \geq j+2, \end{cases} \quad (2)$$

where $B_{l,k} := \prod_{i=l-1}^{k-1} \beta_i p_{i+1} = \prod_{i=l-1}^{k-1} \gamma_i$ (empty product equals 1).

The base $j = 1$ gives $L_1 v_l = (1+q)v_l$ for $l \geq 3$. For the inductive step $j-1 \rightarrow j$: the operator T_{j+1} acts as $(1+q)$ on v_k for $k \neq j, j+1$, and on the block $\text{span}(v_j, v_{j+1})$ as

$$(T_{j+1}+1)v_j = (1+q)p_j u_j, \quad (T_{j+1}+1)v_{j+1} = u_j = v_j + \beta_j v_{j+1},$$

using $q_j = 1/(1+q)$. Thus $(T_{j+1}+1)u_{j-1} = (T_{j+1}+1)(v_{j-1} + \beta_{j-1} v_j) = (1+q)v_{j-1} + (1+q)\beta_{j-1} p_j u_j = (1+q)(v_{j-1} + \gamma_{j-1} u_j)$. Substituting these into the inductive form for L_{j-1} and collecting yields (??).

Reading off the matrix. Apply at $j = n-2$. For source v_l with $3 \leq l \leq n-1$, the image $L_{n-2} v_l$ is supported on the image-basis vectors $v_{l-1}, v_l, \dots, v_{n-3}, u_{n-2}$, with leading coefficient $(1+q)^{n-2} q_{l-1} = (1+q)^{n-3}$ (using $q_{l-1} = 1/(1+q)$) at v_{l-1} . For $l = n-1$: $L_{n-2} v_{n-1} = (1+q)^{n-2} q_{n-2} u_{n-2} = (1+q)^{n-3} u_{n-2}$. For $l = n$: $L_{n-2} v_n = (1+q)^{n-2} v_n$.

In short, the matrix M from source basis $(v_3, v_4, \dots, v_n)$ to image basis $(v_2, v_3, \dots, v_{n-3}, u_{n-2}, v_n)$ has:

- Column for v_l , $3 \leq l \leq n-1$: leading $(1+q)^{n-3}$ at row v_{l-1} (or row u_{n-2} if $l = n-1$); off-diagonal entries $(1+q)^{n-3} \prod_{i=l-1}^{k-1} \gamma_i$ at row v_k (or u_{n-2} for $k = n-2$), for $l \leq k \leq n-2$.

- Column for v_n : $(1+q)^{n-2}$ at row v_n , zero elsewhere.

Reading column-by-column, this is lower-triangular with all diagonal entries equal to $(1+q)^{n-3}$ in the rows above v_n , and $(1+q)^{n-2}$ on the v_n -diagonal.

Back-substitution. Solve $z = Mw$ row by row. Index the source basis as $w_3, w_4, \dots, w_n$ and the image basis as $z^{(v_2)}, z^{(v_3)}, \dots, z^{(v_{n-3})}, z^{(u_{n-2})}, z^{(v_n)}$.

Row v_2 : $z^{(v_2)} = (1+q)^{n-3}w_3$, so $w_3 = z^{(v_2)}/(1+q)^{n-3}$.

Row v_k , $3 \leq k \leq n-3$: $z^{(v_k)} = (1+q)^{n-3} \left[\sum_{l=3}^k \Pi_{l,k} w_l + w_{k+1} \right]$, where $\Pi_{l,k} := \prod_{i=l-1}^{k-1} \gamma_i$.

We claim by induction on k that

$$w_{k+1} = \frac{z^{(v_k)} - \gamma_{k-1} z^{(v_{k-1})}}{(1+q)^{n-3}} \quad (3)$$

for $3 \leq k \leq n-3$ (with $z^{(v_2)}$ taking the place of $z^{(v_{k-1})}$ at $k=3$). The case $k=3$: $z^{(v_3)} = (1+q)^{n-3}[\gamma_2 w_3 + w_4]$, so $w_4 = z^{(v_3)}/(1+q)^{n-3} - \gamma_2 w_3 = (z^{(v_3)} - \gamma_2 z^{(v_2)})/(1+q)^{n-3}$, matching (??).

For the inductive step $k-1 \rightarrow k$: by the inductive hypothesis, $w_{l+1} = (z^{(v_l)} - \gamma_{l-1} z^{(v_{l-1})})/(1+q)^{n-3}$ for $l < k$. Substituting into the row- v_k equation and using the telescoping identity $\Pi_{l,k} = \gamma_{l-1} \Pi_{l-1,k}$ – which follows directly from $\Pi_{l,k} = \prod_{i=l-1}^{k-1} \gamma_i$ – all terms with $z^{(v_l)}$ for $l < k-1$ cancel, leaving exactly (??). Specifically, after substituting, the contribution of $z^{(v_l)}$ to w_{k+1} is

$$\left[-\Pi_{l+1,k} + \Pi_{l+2,k} \gamma_l \right] \cdot \frac{z^{(v_l)}}{(1+q)^{n-3}} = \left[-\Pi_{l+1,k} + \Pi_{l+1,k} \right] \cdot \frac{z^{(v_l)}}{(1+q)^{n-3}} = 0$$

for $2 \leq l \leq k-2$ (using $\Pi_{l+2,k} \gamma_l = \Pi_{l+1,k}$), while the $z^{(v_{k-1})}$ contribution survives as $-\gamma_{k-1} z^{(v_{k-1})}/(1+q)^{n-3}$.

Row u_{n-2} : $z^{(u_{n-2})} = (1+q)^{n-3} \left[\sum_{l=3}^{n-2} \Pi_{l,n-2} w_l + w_{n-1} \right]$. By an identical telescoping argument, all $z^{(v_l)}$ contributions for $l \leq n-4$ cancel, leaving

$$w_{n-1} = \frac{z^{(u_{n-2})} - \gamma_{n-3} z^{(v_{n-3})}}{(1+q)^{n-3}}. \quad (4)$$

Translating to y . The element $y \in \text{span}(v_2, \dots, v_{n-2})$ relates to z via the constraint $(T_{n-1}+1)z = y$. Since y has no v_{n-1} or v_n component, and $T_{n-1}+1$ has eigenvalues $1+q$ on $\text{span}(v_2, \dots, v_{n-2})$ and 0 on $\text{span}(u'_{n-1})$:

$$z = \frac{1}{1+q} y + \mu u'_{n-1}$$

for some $\mu \in \mathbb{Q}(q)$ chosen so that $z \in I_{n-2}$. Working out the constraint (May-8 evening, eq. (2.2)) yields

$$z^{(v_l)} = y_{v_l}/(1+q), \quad 2 \leq l \leq n-3, \quad z^{(u_{n-2})} = y_{v_{n-2}}/((1+q)\alpha_{n-2}) = y_{v_{n-2}}/(1+q),$$

using $\alpha_{n-2} = 1$.

Substituting into (??) and (??): for $3 \leq k \leq n-3$,

$$w_{k+1} = \frac{y_{v_k}/(1+q) - \gamma_{k-1} y_{v_{k-1}}/(1+q)}{(1+q)^{n-3}} = \frac{y_{v_k} - \gamma_{k-1} y_{v_{k-1}}}{(1+q)^{n-2}};$$

and for w_{n-1} :

$$w_{n-1} = \frac{y_{v_{n-2}}/(1+q) - \gamma_{n-3} y_{v_{n-3}}/(1+q)}{(1+q)^{n-3}} = \frac{y_{v_{n-2}} - \gamma_{n-3} y_{v_{n-3}}}{(1+q)^{n-2}}.$$

Setting $l = k + 1$ in the first and matching the second: (??) is established for all $3 \leq l \leq n - 1$.

The case $l = n$ (Image Lemma) was proved May-8 evening. $\square$

Remark 2. The Strong Image formula (??) extends the Image Lemma from w_{v_n} (which depends only on $y_{v_{n-2}}$) to every “intermediate” coordinate w_{v_l} for $l = 3, \dots, n - 1$, each of which depends on *exactly two* consecutive coordinates of y .

This local structure is the key technical ingredient in what follows: it converts the question of (Q_m) on K_m into a linear-functional non-vanishing question on K_{m-1} .

3 Slopes on the new generators

We now use the Strong Image formula to extract two “slopes” on the new-generator quotient of K_l , indexed by parity.

Definition 3 (Slopes μ_l and ν_l). For $l \geq 4$, let $\xi^{(l)} \in K_l$ be any vector with $\xi^{(l)} \notin K_{l-2}^{\text{embed}}$ (such a vector exists since K_{l-2}^{embed} has codimension 1 in K_l).

- For l **odd** ($l \geq 5$): $\xi_{v_l}^{(l)} = 0$ by (P_l) and $\xi_{v_{l-1}}^{(l)} \neq 0$ by (Q_l) (the latter is automatic from $(P_l), (P_{l-1})$ as shown in May-9 §2). Define

$$\mu_l := \frac{\xi_{v_{l-1}}^{(l)}}{\xi_{v_{l-2}}^{(l)}} \quad (\text{when } \xi_{v_{l-2}}^{(l)} \neq 0).$$

- For l **even** ($l \geq 4$): $\xi_{v_l}^{(l)} \neq 0$ by (P_l) . Define

$$\nu_l := \frac{\xi_{v_{l-1}}^{(l)}}{\xi_{v_l}^{(l)}}.$$

Both are independent of the choice of $\xi^{(l)}$ within its coset modulo K_{l-2}^{embed} (since $K_{l-2}^{\text{embed}} \subset \text{span}(v_2, \dots, v_{l-2})$ has zero v_{l-1}, v_l components).

Lemma 4 (Base values). $\nu_4 = -1/\gamma_2$ and $\mu_5 = -\gamma_2$.

Proof. $K_4 = \text{span}(v_2, v_3 - cv_4)$ with $c = \gamma_2$ (verified in May-9 nesting paper, via $c = q_2\beta_2p_3/q_3$). The new generator $\eta^{(4)} = v_3 - \gamma_2v_4$ gives $\nu_4 = 1/(-\gamma_2) = -1/\gamma_2$.

For μ_5 : $K_5 = \text{span}(v_2, v_3 - cv_4)$ as well (since (P_5) for 5 odd gives $v_5^*(K_5) = 0$, and the dim and $K_4^{\text{embed}} \subset K_5$ force equality). So $\xi^{(5)} = v_3 - \gamma_2v_4$ (taken modulo $\mathbb{Q}(q)v_2 = K_3^{\text{embed}}$), giving $\mu_5 = -\gamma_2/1 = -\gamma_2$. $\square$

Lemma 5 (Recursion for μ_l , l odd ≥ 7). Assume (R_{l-1}) holds at level $l - 1$ even. Then

$$\mu_l = \frac{-\gamma_{l-3}}{1 - \gamma_{l-4} \nu_{l-3}}. \tag{5}$$

Proof. Pick $\xi^{(l)} \in K_l$ new generator. By the May-9 reduction analysis at level l : $R'_l(\xi^{(l)}) = y$ lies in $D_l \cap K_{l-1}^{\text{embed}} = K_{l-1}^{\text{embed}} \cap \text{span}(v_2, \dots, v_{l-2})$. By (P_{l-1}) for $l - 1$ even, $v_{l-1}^*(K_{l-1}) \neq 0$ but we additionally need to know the structure of $K_{l-1} \cap \{v_{l-1}^* = 0\}$.

By (R_{l-1}) assumed: v_{l-2}^* and v_{l-1}^* are proportional on K_{l-1} with nonzero ratio; hence $K_{l-1} \cap \{v_{l-1}^* = 0\} = K_{l-1} \cap \{v_{l-2}^* = 0\}$, and by nesting + dimension this equals K_{l-3}^{embed} (in $V_{(l-2,1)}$).

So $y \in K_{l-3}^{\text{embed}}$, which has support on $v_2, \dots, v_{l-3}$ since $K_{l-3} \subset V_{(l-4,1)}$ (natural embedding). In particular, $y_{v_{l-3}}$ may be nonzero while $y_{v_{l-2}} = 0$.

By Strong Image at level $n = l$:

$$\begin{aligned}\xi_{v_{l-1}}^{(l)} &= \frac{y_{v_{l-2}} - \gamma_{l-3} y_{v_{l-3}}}{(1+q)^{l-2}} = \frac{-\gamma_{l-3} y_{v_{l-3}}}{(1+q)^{l-2}}, \\ \xi_{v_{l-2}}^{(l)} &= \frac{y_{v_{l-3}} - \gamma_{l-4} y_{v_{l-4}}}{(1+q)^{l-2}}.\end{aligned}$$

Hence

$$\mu_l = \frac{\xi_{v_{l-1}}^{(l)}}{\xi_{v_{l-2}}^{(l)}} = \frac{-\gamma_{l-3} y_{v_{l-3}}}{y_{v_{l-3}} - \gamma_{l-4} y_{v_{l-4}}} = \frac{-\gamma_{l-3}}{1 - \gamma_{l-4} (y_{v_{l-4}}/y_{v_{l-3}})}.$$

Finally the ratio $y_{v_{l-4}}/y_{v_{l-3}}$ on K_{l-3}^{embed} : since $K_{l-3} = K_{l-5}^{\text{embed}} \oplus \mathbb{Q}(q)\eta^{(l-3)}$ and $K_{l-5}^{\text{embed}} \subset \text{span}(v_2, \dots, v_{l-5})$ (both v_{l-4}, v_{l-3} coordinates vanish on K_{l-5}^{embed}), the ratio equals $\eta_{v_{l-4}}^{(l-3)}/\eta_{v_{l-3}}^{(l-3)} = \nu_{l-3}$. Substituting gives (??). $\square$

Lemma 6 (Recursion for ν_l , l even ≥ 6).

$$\nu_l = \frac{1 - \gamma_{l-3}/\mu_{l-1}}{(1+q)^{l-2} C_l}, \quad C_l = \frac{\beta_{l-2}\beta'_{l-1}}{(1+q)^{l-1}}. \quad (6)$$

Proof. Pick $\eta^{(l)} \in K_l$ new generator. Then $R_l'(\eta^{(l)}) = z \in K_{l-1}^{\text{embed}}$ where K_{l-1} is at level $l-1$ odd, so z has support on $\text{span}(v_2, \dots, v_{l-2})$ (by (P_{l-1})). By Image Lemma: $\eta_{v_l}^{(l)} = C_l z_{v_{l-2}}$. Since $\eta^{(l)} \notin K_{l-2}^{\text{embed}}$ implies $\eta_{v_l}^{(l)} \neq 0$, we have $z_{v_{l-2}} \neq 0$.

By Strong Image: $\eta_{v_{l-1}}^{(l)} = (z_{v_{l-2}} - \gamma_{l-3} z_{v_{l-3}})/(1+q)^{l-2}$. So

$$\nu_l = \frac{\eta_{v_{l-1}}^{(l)}}{\eta_{v_l}^{(l)}} = \frac{z_{v_{l-2}} - \gamma_{l-3} z_{v_{l-3}}}{(1+q)^{l-2} C_l z_{v_{l-2}}} = \frac{1 - \gamma_{l-3} (z_{v_{l-3}}/z_{v_{l-2}})}{(1+q)^{l-2} C_l}.$$

The ratio $z_{v_{l-3}}/z_{v_{l-2}}$ on K_{l-1} (odd): since $K_{l-1} = K_{l-3}^{\text{embed}} \oplus \mathbb{Q}(q)\xi^{(l-1)}$ and $K_{l-3}^{\text{embed}} \subset \text{span}(v_2, \dots, v_{l-4})$ (the latter inclusion since $K_{l-3} \subset V_{(l-4,1)}$ embeds as $\text{span}(v_2, \dots, v_{l-4})$ in $V_{(l-2,1)}$, with no v_{l-3}, v_{l-2} components), the ratio is determined by $\xi^{(l-1)}$: $z_{v_{l-3}}/z_{v_{l-2}} = \xi_{v_{l-3}}^{(l-1)}/\xi_{v_{l-2}}^{(l-1)} = 1/\mu_{l-1}$.

Substituting yields (??). $\square$

4 Sign induction

We now prove an alternating sign property of the slopes at any positive real q , treating $q > 0$ as a real variable in $\mathbb{R}_{>0}$.

Lemma 7 (Positivity of Hoefsmit constants). *At any real $q > 0$, the constants $\beta_i, p_i, q_i, \gamma_i$ are strictly positive, and β'_i is strictly negative. In particular, $C_l = \beta_{l-2}\beta'_{l-1}/(1+q)^{l-1} < 0$.*

Proof. $[k] = (q^k - 1)/(q - 1) > 0$ at $q > 0, q \neq 1$, and $[k]|_{q=1} = k > 0$. Therefore $\beta_i = q[i-1]/[i] > 0$, $p_i = [i+1]/([2][i]) > 0$, $q_i = 1/[2] > 0$, and $\gamma_i = \beta_i p_{i+1} > 0$. $\beta'_i = -[i+1]/[i] < 0$. Hence $C_l = (\text{positive}) \cdot (\text{negative})/(\text{positive}) < 0$. $\square$

Theorem 8 (Alternating sign). *For all real $q > 0$:*

- $\nu_l < 0$ for every even $l \geq 4$.
- $\mu_l < 0$ for every odd $l \geq 5$.

The proof is by strong induction simultaneously establishing $(P_l), (Q_l), (R_l)$ along with these signs.

Proof. Order the levels $4 < 5 < 6 < 7 < \dots$ and induct on l .

Bases.

- Level 3: $K_3 = \mathbb{Q}(q)v_2$, $(P_3), (Q_3)$ trivial.
- Level 4: $K_4 = \text{span}(v_2, v_3 - \gamma_2 v_4)$ (computed). $(P_4), (Q_4), (R_4)$ all hold. $\nu_4 = -1/\gamma_2 < 0$ by Lemma ??.
- Level 5: $K_5 = \text{span}(v_2, v_3 - \gamma_2 v_4)$ (extended by zero on v_5). $(P_5), (Q_5)$ hold, $\mu_5 = -\gamma_2 < 0$.

Inductive step. Suppose all properties (sign, (P) , (Q) , (R)) hold at all levels $< l$.

Case l odd, $l \geq 7$. By induction, (R_{l-1}) holds (since $l-1$ even is $< l$, and (R_{l-1}) follows from $(P_{l-1}), (Q_{l-1})$ via May-9 Reduction Theorem). Apply Lemma ??: $\mu_l = -\gamma_{l-3}/(1 - \gamma_{l-4}\nu_{l-3})$.

By induction, $\nu_{l-3} < 0$ (since $l-3$ even is $< l$). By Lemma ??, $\gamma_{l-4} > 0$. Hence $\gamma_{l-4}\nu_{l-3} < 0$, so $1 - \gamma_{l-4}\nu_{l-3} > 1 > 0$. Therefore

$$\mu_l = \frac{-\gamma_{l-3}}{\text{positive}} < 0.$$

We need to check that $\xi_{v_{l-2}}^{(l)} \neq 0$ (so that μ_l is finite). By Lemma ??'s computation, $\xi_{v_{l-2}}^{(l)} = (y_{v_{l-3}}/(1+q)^{l-2})(1 - \gamma_{l-4}\nu_{l-3})$, which is nonzero since both factors are.

Properties $(P_l), (Q_l)$ (with (Q_l) for l odd automatic): established via the May-8 evening induction from $(P_{l'}), (R_{l'})$ at smaller l' .

Case l even, $l \geq 6$. By induction, $\mu_{l-1} < 0$ (since $l-1$ odd is $< l$, with $(P_{l-1}), (Q_{l-1})$ valid). Apply Lemma ??: $\nu_l = (1 - \gamma_{l-3}/\mu_{l-1})/((1+q)^{l-2}C_l)$.

By induction, $\mu_{l-1} < 0$, $\gamma_{l-3} > 0$, hence $\gamma_{l-3}/\mu_{l-1} < 0$, so $1 - \gamma_{l-3}/\mu_{l-1} > 1 > 0$. By Lemma ??, $C_l < 0$, so $(1+q)^{l-2}C_l < 0$. Therefore

$$\nu_l = \frac{\text{positive}}{\text{negative}} < 0.$$

For (P_l) at level l even: by May-8 evening induction, $(P_l) \Leftarrow (P_{l-1}), (Q_{l-1})$ (which we have).

For (Q_l) at level l even: see Theorem ?? below.

For (R_l) at level l even: by May-9 Reduction Theorem, $(R_l) \Leftarrow (P_l), (Q_l)$, Nesting, all of which we have. $\square$

5 The main theorem

Theorem 9 (Property (Q_m) for all even m). *For every even $m \geq 4$, the linear functional v_{m-1}^* is not identically zero on K_m . Equivalently, there exists $w \in K_m$ with $w_{v_{m-1}} \neq 0$.*

Proof. The case $m = 4$ is base: $K_4 = \text{span}(v_2, v_3 - \gamma_2 v_4)$ contains $v_3 - \gamma_2 v_4$ with v_3 -coordinate 1 $\neq 0$.

For $m \geq 6$ even: by Theorem ?? (sign induction established at all levels $< m$), $\mu_{m-1} < 0$ at every $q > 0$. By Lemma ??, $\gamma_{m-3} > 0$. Therefore

$$\mu_{m-1} - \gamma_{m-3} < 0$$

at every $q > 0$. Since $\mu_{m-1} - \gamma_{m-3}$ is a rational function in q that is strictly negative at every positive real q , it cannot be identically zero. Hence $\mu_{m-1} - \gamma_{m-3} \neq 0$ in $\mathbb{Q}(q)$.

Pick $\xi^{(m-1)} \in K_{m-1}$ a new generator of K_{m-1} with $\xi_{v_{m-2}}^{(m-1)} \neq 0$ (exists by (Q_{m-1}) , automatic for $m-1$ odd). By Theorem ??'s argument, $\xi_{v_{m-3}}^{(m-1)} \neq 0$ as well. Embed $\xi^{(m-1)}$ into $V_{(m-1,1)}$ via the natural embedding, and consider the linear functional

$$\phi_m := v_{m-2}^* - \gamma_{m-3} v_{m-3}^*$$

applied to $\xi^{(m-1)}$:

$$\phi_m(\xi^{(m-1)}) = \xi_{v_{m-2}}^{(m-1)} - \gamma_{m-3} \xi_{v_{m-3}}^{(m-1)} = \xi_{v_{m-3}}^{(m-1)} (\mu_{m-1} - \gamma_{m-3}) \neq 0.$$

Now lift $\xi^{(m-1)}$ to $w \in K_m$ via $R_m'^{-1}$: pick any $w \in V_{(m-1,1)}$ with $R_m'(w) = \xi^{(m-1)}$. By the May-8 evening Reduction Theorem (using the Image Lemma and K_{m-1} analysis), such w lies in K_m provided $\xi^{(m-1)} \in D_m \cap K_{m-1}^{\text{embed}}$. Since $\xi_{v_{m-1}}^{(m-1)} = 0$ ((P_{m-1}) for $m-1$ odd), $\xi^{(m-1)} \in \text{span}(v_2, \dots, v_{m-2}) \subset D_m$. And by the inductive hypothesis $\xi^{(m-1)} \in K_{m-1}^{\text{embed}}$. So $w \in K_m$.

By Strong Image at level $n = m$ applied to w with $y = \xi^{(m-1)}$:

$$w_{v_{m-1}} = \frac{\xi_{v_{m-2}}^{(m-1)} - \gamma_{m-3} \xi_{v_{m-3}}^{(m-1)}}{(1+q)^{m-2}} = \frac{\phi_m(\xi^{(m-1)})}{(1+q)^{m-2}} \neq 0.$$

This proves (Q_m) . □

Corollary 10 (Hook rank formula, unconditional). *For every $n \geq 2$,*

$$\text{rank } \Pi^{S_n}|_{V_{(n-1,1)}} = \lceil \frac{n-2}{2} \rceil.$$

Proof. By Theorem ??, (Q_m) holds for every even $m \geq 4$. Combined with the May-9 Reduction Theorem and the rank-formula proof in May-9 §4, the rank formula holds for all $n \geq 2$. □

6 Computational verification

The Strong Image Coefficient Formula (??) is verified symbolically at $n = 4, 5, 6, 7, 8$ in `verify_strong_image.py`, where the formula is matched against direct linear-algebra computation of w given y .

The slopes μ_l, ν_l have been verified explicitly:

- $\nu_4 = -1/\gamma_2 = -(q+1)(q^2+q+1)/(q(q^2+1))$.
- $\mu_5 = -\gamma_2 = -q(q^2+1)/((q+1)(q^2+q+1))$.
- $\nu_6 =$ (computed in May-8 evening Table).
- $\mu_7 = -\gamma_2\gamma_4/(\gamma_2+\gamma_3) = -q(q^2+1)(q^2-q+1)(q^2+q+1)/((q+1)(2q^2-q+2)(q^4+q^3+q^2+q+1))$.

The identity $\mu_{m-1} - \gamma_{m-3}$ for (Q_m) at $m = 6, 8$:

- $m = 6$: $\mu_5 - \gamma_3 = -(\gamma_2 + \gamma_3) = -q(2q^2 - q + 2)/((q+1)(q^2+1))$, nonzero.
- $m = 8$: $\mu_7 - \gamma_5 = -q(q^2+1)(3q^4-2q^3+4q^2-2q+3)/((q+1)(q^2-q+1)(q^2+q+1)(2q^2-q+2))$, nonzero.

Both negative at $q > 0$ as predicted by Theorem ??.

7 Honest assessment

What this paper proves.

1. Theorem ?? (Strong Image Coefficient Formula). Rigorous via direct triangular matrix inversion of R''_{n-1} on $V_{(n-1,1)}$.
2. Lemmas ?? and ??: recursive formulas for the slopes μ_l, ν_l .
3. Theorem ?? (alternating sign): both slopes strictly negative at every positive real q .
4. Theorem ??: (Q_m) holds for all even $m \geq 4$.
5. Corollary ??: hook rank formula proved unconditionally.

Where Hoefsmit-specific computation enters. The proof crucially uses positivity of the explicit Hoefsmit constants β_i, p_i, γ_i at $q > 0$, and negativity of β'_i . These hold because $V_{(n-1,1)}$'s seminormal block matrices have a uniform sign structure under our normalization.

Inductive scope. The induction proceeds simultaneously on $(P_l), (Q_l), (R_l)$ for $l \leq m$, with the sign properties of μ_l, ν_l as additional inductive invariants. Each level requires properties at all strictly smaller levels. No level uses its own (R_l) or (Q_l) in deriving the next level.

Files.

- `~/projects/proofs/2026-05-10-Q-m-via-strong-image.tex` (this writeup).
- `~/projects/scratch/2026-05-09-nesting-identity/check_identity.py` (Hoefsmit constant values).
- `~/projects/scratch/2026-05-08-kernel-dichotomy/check_n8_proportionality.py` (computational verification of $(Q_m), (R_m)$ at $m \leq 8$).

What remains. With (Q_m) proved, the May-9 Reduction Theorem closes the hook rank formula unconditionally. Future directions:

- Closed form for the μ_l, ν_l progressions (continued fraction in the Hoefsmit gammas).
- Generalization to other 2-row partitions $(n - r, r)$ for $r \geq 2$.
- Cellular interpretation: does the alternating sign reflect a duality between the seminormal form and another preferred basis (Kazhdan-Lusztig, web bases, etc.)?